

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0454

Orange 138 kV
March 23, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0454. The Interconnection Customer proposes a 81.6 MW solar photovoltaic facility interconnecting at the Orange 138 kV substation in the New Jersey area. The request was received on April 01, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 271 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$47,395,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
WHITE ROCK-NIKE – MAGNOLIA 1708 138kV	CRAWFORDSVILLE POWER PLANT – PINE TREE ACRES 138kV L	250.7	217	7.0
BABINE LAKE – DAM HEIGHTS 138kV Line	WHITE ROCK-NIKE – MAGNOLIA 1708 138kV line	227.6	226	8.3
MULTITRADE OF PITTSYLVANIA LP – SNE - SANSUM N ELEC 138kV	MAGOLIA 1708 – WHITE ROCK-NIKE 138kV	223.5	262	12.4
JAN PHYL – SNE - SANSUM N ELEC SUBSTATION 138kV	MAGNOLIA 1708 – WHITE ROCK-NIKE 138kV	215.1	257	12.4
BABINE LAKE – JAN PHYL 138kV Line	WHITE ROCK-NIKE – MAGNOLIA 1708 138kV Line	208.6	236	12.4
NORTHEASTERN – MULTITRADE OF PITTSYLVANIA LP 138kV	CEDAR GROVE – VEVE CALZADA 2002 138kV	192.0	244	11.5
MAGNOLIA 1708 – WEST ORANGE 138kV Line	SNE - SANSU N ELEC – MULTITRADE OF PITTSYLVANIA LP 138kV line	181.9	238	19.5
CEDAR GROVE – VEVE CALZADA 2002 138kV	MULTITRADE OF PITTSYLVANIA LP – NORTHEASTERN 138kV Line	181.3	260	5.1
CEDAR GROVE – MULTITRADE OF PITTSYLVANIA LP 138kV Line	NORTHEASTERN – MULTITRADE OF PITTSYLVANIA LP 138kV	172.7	251	5.1
WHIPPANY – ORANGE 138kV L	MARION DRIVE – WHITE ROCK 138kV	154.2	258	19.7
MT POSO – WHIPPANY 138kV line	MARION DRIVE – WHITE ROCK SUBSTATION 138kV	146.1	249	19.7
MT POSO – W ORANGE 138kV	MARION DRIVE – WHITE ROCK 138kV	136.6	226	19.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SUNTREE – ORANGE 138kV	WHIPPANY – ORANGE 138kV line	132.2	251	32.8
W CALDWELL – SUNTREE 138kV	WHIPPANY – ORANGE 138kV	128.8	243	32.8
NORTHEASTERN – WEST CALDWELL 138kV	WHIPPANY – ORANGE 138kV	122.0	229	32.8
ATHENIA – DAM HEIGHTS 138kV	WASCO – ARTESIAN 138kV	118.6	214	5.1
LOCUST GROVE – SHREDDER 138kV L	MAGNOLIA 1708 – WHITE ROCK-NIKE SUBSTATION 138kV L	115.7	289	7.4
WHITE ROCK-NIKE – 15TH ST SUBSTATION 138kV Line	SNE - SANSUM N ELEC – MULTITRADE OF PITTSYLVANIA LP 138kV	106.2	253	8.2
ARTESIAN – DAM HEIGHTS 138kV	W ORANGE – MAGNOLIA 1708 138kV L	105.3	228	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHITE ROCK-NIKE – MAGNOLIA 1708 138kV	Rebuild line with higher-capacity conductor	\$1,641,000	100% (\$1,641,000)
DAM HEIGHTS – BABINE LAKE 138kV	Rebuild line with higher-capacity conductor	\$997,000	100% (\$997,000)
MULTITRADE OF PITTSYLVANIA LP – SNE - SANSUM N ELEC 138kV	Rebuild line with higher-capacity conductor	\$1,132,000	100% (\$1,132,000)
JAN PHYL – SNE - SANSUM N ELEC SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$751,000	100% (\$751,000)
BABINE LAKE – JAN PHYL 138kV Line	Rebuild line with higher-capacity conductor	\$1,234,000	100% (\$1,234,000)
MULTITRADE OF PITTSYLVANIA LP – NORTHEASTERN 138kV Line	Rebuild line with higher-capacity conductor	\$4,851,000	100% (\$4,851,000)
MAGNOLIA 1708 – W ORANGE 138kV	Rebuild line with higher-capacity conductor	\$961,000	100% (\$961,000)
CEDAR GROVE – VEVE CALZADA 2002 138kV	Rebuild line with higher-capacity conductor	\$7,051,000	100% (\$7,051,000)
CEDAR GROVE – MULTITRADE OF PITTSYLVANIA LP 138kV Line	Rebuild line with higher-capacity conductor	\$1,920,000	100% (\$1,920,000)
WHIPPANY – ORANGE 138kV L	Rebuild line with higher-capacity conductor	\$3,487,000	100% (\$3,487,000)
MT POSO – WHIPPANY 138kV line	Rebuild line with higher-capacity conductor	\$6,051,000	100% (\$6,051,000)
MT POSO – W ORANGE 138kV	Rebuild line with higher-capacity conductor	\$4,780,000	100% (\$4,780,000)
SUNTREE – ORANGE 138kV	Rebuild line with higher-capacity conductor	\$1,187,000	100% (\$1,187,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W CALDWELL – SUNTREE 138kV	Rebuild line with higher-capacity conductor	\$3,444,000	100% (\$3,444,000)
NORTHEASTERN – WEST CALDWELL 138kV	Rebuild line with higher-capacity conductor	\$3,157,000	100% (\$3,157,000)
ATHENIA – DAM HEIGHTS 138kV	Rebuild line with higher-capacity conductor	\$2,293,000	100% (\$2,293,000)
SHRDDER – LOCUST GROVE 138kV L	Rebuild line with higher-capacity conductor	\$939,000	100% (\$939,000)
WHITE ROCK-NIKE – 15TH ST SUBSTATION 138kV Line	Reconductor line	\$691,000	100% (\$691,000)
ARTESIAN – DAM HTS 138kV line	Reconductor line	\$828,000	100% (\$828,000)
	Total allocated to project		\$47,395,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.